

QUBO Formulation: Dynamic Spectrum Allocation Problem

Dynamic Spectrum Allocation Problem

Problem Statement. Given a set of N_L radio links and N_C available channels, the Dynamic Spectrum Allocation Problem requires assigning exactly one channel to each link. Each link-channel pair (l, c) yields a utility score $U_{l,c}$. A subset of link pairs $\mathcal{I} \subseteq \{0, \dots, N_L - 1\}^2$ are designated as interfering; if two interfering links l_1 and l_2 are assigned the same channel c , a penalty P_{l_1, l_2} is incurred. The task is to find a channel assignment that maximizes the total utility minus the total interference penalty.

Decision Variables. For each link $l \in \{0, \dots, N_L - 1\}$ and channel $c \in \{0, \dots, N_C - 1\}$, define a binary decision variable $x_{l,c} \in \{0, 1\}$. The variable $x_{l,c}$ equals 1 if and only if link l is assigned channel c , and 0 otherwise.

QUBO Derivation. Step 1 – Objective Conversion. The original goal is to maximize utility minus interference. Converting to a minimization problem suitable for QUBO yields:

$$\mathcal{H}_{\text{obj}}(x) = - \sum_{l=0}^{N_L-1} \sum_{c=0}^{N_C-1} U_{l,c} x_{l,c} + \sum_{(l_1, l_2) \in \mathcal{I}} \sum_{c=0}^{N_C-1} P_{l_1, l_2} x_{l_1, c} x_{l_2, c}. \quad (1)$$

Step 2 – Constraint Formulation. Each link must be assigned exactly one channel. This is enforced by the equality constraints:

$$\sum_{c=0}^{N_C-1} x_{l,c} = 1, \quad \forall l \in \{0, \dots, N_L - 1\}. \quad (2)$$

Step 3 – Penalty Expansion. We introduce a penalty weight $\lambda > 0$ and convert each constraint into a quadratic penalty term. For a fixed link l , the penalty is:

$$\mathcal{P}_l(x) = \lambda \left(\sum_{c=0}^{N_C-1} x_{l,c} - 1 \right)^2. \quad (3)$$

Expanding the square and applying the idempotent property $x_{l,c}^2 = x_{l,c}$ for binary variables gives:

$$\begin{aligned}
\left(\sum_c x_{l,c} - 1\right)^2 &= \sum_{c=0}^{N_C-1} x_{l,c}^2 + \sum_{c \neq c'} x_{l,c} x_{l,c'} - 2 \sum_{c=0}^{N_C-1} x_{l,c} + 1 \\
&= \sum_{c=0}^{N_C-1} x_{l,c} + 2 \sum_{0 \leq c < c' \leq N_C-1} x_{l,c} x_{l,c'} - 2 \sum_{c=0}^{N_C-1} x_{l,c} + 1 \\
&= 1 - \sum_{c=0}^{N_C-1} x_{l,c} + 2 \sum_{0 \leq c < c' \leq N_C-1} x_{l,c} x_{l,c'}.
\end{aligned} \tag{4}$$

Multiplying by λ yields the general penalty contribution for link l :

$$\mathcal{P}_l(x) = \lambda - \lambda \sum_{c=0}^{N_C-1} x_{l,c} + 2\lambda \sum_{0 \leq c < c' \leq N_C-1} x_{l,c} x_{l,c'}. \tag{5}$$

Step 4 – Combined Hamiltonian. The total QUBO Hamiltonian is the sum of the objective and all link penalties:

$$\mathcal{H}(x) = \mathcal{H}_{\text{obj}}(x) + \sum_{l=0}^{N_L-1} \mathcal{P}_l(x). \tag{6}$$

Step 5 – Q Matrix and Offset. Collecting coefficients from $\mathcal{H}(x)$, the QUBO parameters are defined as follows. The constant offset is $c = \lambda N_L$. The diagonal and off-diagonal entries of the upper-triangular Q matrix are:

$$\begin{aligned}
Q_{(l,c),(l,c)} &= -U_{l,c} - \lambda, \quad \forall l \in \{0, \dots, N_L - 1\}, c \in \{0, \dots, N_C - 1\}, \\
Q_{(l_1,c),(l_2,c)} &= P_{l_1,l_2}, \quad \forall (l_1, l_2) \in \mathcal{I}, c \in \{0, \dots, N_C - 1\}, \\
Q_{(l,c),(l,c')} &= 2\lambda, \quad \forall l \in \{0, \dots, N_L - 1\}, 0 \leq c < c' \leq N_C - 1, \\
Q_{i,j} &= 0, \quad \text{otherwise.}
\end{aligned} \tag{7}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that violating the one-hot constraint is strictly more costly than any potential gain in the objective function. Specifically, we require:

$$\lambda > \max \left(\max_{l,c} |U_{l,c}|, \max_{(l_1,l_2) \in \mathcal{I}} P_{l_1,l_2} \right). \tag{8}$$

If a link l violates its constraint, the penalty term $\mathcal{P}_l(x)$ increases by at least λ . The maximum possible reduction in $\mathcal{H}_{\text{obj}}$ achievable by reassigning channels is bounded by $\max_{l,c} |U_{l,c}|$ (from utilities) or $\max_{(l_1,l_2) \in \mathcal{I}} P_{l_1,l_2}$ (from interference penalties). Since λ exceeds both bounds, any infeasible configuration has an energy strictly greater than the best feasible configuration, guaranteeing that the constraint is respected at the optimum.

Correctness Argument. Minimizing $\mathcal{H}(x)$ over $\{0, 1\}^{N_L N_C}$ recovers the optimal channel assignment because (i) for any feasible assignment, all penalty terms vanish, reducing $\mathcal{H}(x)$ to $\mathcal{H}_{\text{obj}}(x)$; thus, the minimum over feasible points corresponds exactly to the maximum utility minus interference. (ii) For any infeasible assignment, at least one constraint is violated, adding a penalty of at least λ to the energy. By the choice of λ , this penalty strictly dominates any possible decrease in $\mathcal{H}_{\text{obj}}$ resulting from the infeasibility. Consequently, the global minimum of $\mathcal{H}(x)$ must occur at a feasible point that optimizes the original objective function.

Illustrative Example. Consider a concrete instance with $N_L = 2$ links, $N_C = 3$ channels, and a single interfering pair $\mathcal{I} = \{(0, 1)\}$. The decision variables are mapped to indices $i = l \cdot N_C + c$, yielding $x_{0,0} \rightarrow 0$, $x_{0,1} \rightarrow 1$, $x_{0,2} \rightarrow 2$, $x_{1,0} \rightarrow 3$, $x_{1,1} \rightarrow 4$, $x_{1,2} \rightarrow 5$. Substituting these indices into the general formulas yields the concrete objective:

$$\begin{aligned} \mathcal{H}_{\text{obj}}(x) = & -U_{0,0}x_{0,0} - U_{0,1}x_{0,1} - U_{0,2}x_{0,2} - U_{1,0}x_{1,0} - U_{1,1}x_{1,1} - U_{1,2}x_{1,2} \\ & + P_{0,1}(x_{0,0}x_{1,0} + x_{0,1}x_{1,1} + x_{0,2}x_{1,2}). \end{aligned} \quad (9)$$

The penalty terms for $l = 0$ and $l = 1$ expand to:

$$\begin{aligned} \mathcal{P}_0(x) &= \lambda - \lambda(x_{0,0} + x_{0,1} + x_{0,2}) + 2\lambda(x_{0,0}x_{0,1} + x_{0,0}x_{0,2} + x_{0,1}x_{0,2}), \\ \mathcal{P}_1(x) &= \lambda - \lambda(x_{1,0} + x_{1,1} + x_{1,2}) + 2\lambda(x_{1,0}x_{1,1} + x_{1,0}x_{1,2} + x_{1,1}x_{1,2}). \end{aligned} \quad (10)$$

Combining these, the full Hamiltonian $\mathcal{H}(x) = \mathcal{H}_{\text{obj}}(x) + \mathcal{P}_0(x) + \mathcal{P}_1(x)$ has the following Q matrix entries and offset:

$$\begin{aligned} Q_{0,0} &= -U_{0,0} - \lambda, & Q_{1,1} &= -U_{0,1} - \lambda, & Q_{2,2} &= -U_{0,2} - \lambda, \\ Q_{3,3} &= -U_{1,0} - \lambda, & Q_{4,4} &= -U_{1,1} - \lambda, & Q_{5,5} &= -U_{1,2} - \lambda, \\ Q_{0,3} &= P_{0,1}, & Q_{1,4} &= P_{0,1}, & Q_{2,5} &= P_{0,1}, \\ Q_{0,1} &= 2\lambda, & Q_{0,2} &= 2\lambda, & Q_{1,2} &= 2\lambda, \\ Q_{3,4} &= 2\lambda, & Q_{3,5} &= 2\lambda, & Q_{4,5} &= 2\lambda, \\ \text{offset } c &= 2\lambda. \end{aligned} \quad (11)$$

For a specific numeric instantiation, let $U_{l,c} = 5$ for all l, c , $P_{0,1} = 2$, and $\lambda = 10$. The objective coefficients become -5 , interference terms 2 , and penalty terms -10 (diagonal) and 20 (off-diagonal). The constant offset is 20 . Any valid assignment (e.g., $x_{0,0} = 1, x_{1,1} = 1$, all others 0) yields an energy of -10 . Assigning both links to channel 0 (e.g., $x_{0,0} = 1, x_{1,0} = 1$) violates the one-hot constraint and incurs a penalty of 10 , resulting in an energy of $-10 + 10 = 0$. Thus, the global minimum is -10 , achieved by any feasible assignment where the two links use different channels, correctly maximizing utility while avoiding interference.